

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)	Substitution $n = \frac{pV}{RT} = \frac{(5 \times 10^5)(8.5 \times 10^{-3})}{(8.31)(285)}$ (1) $= 1.79$ [mol] (1)	1	1		2	2	
		(ii)	$N = N_A n = (6.02 \times 10^{23}) \times 1.79$ ecf $= 1.08 \times 10^{24}$		1		1	1	
		(iii)	Substitution of p and V or k and T (1) Correct use of Nm or m in either: $p = \frac{1}{3} \rho \overline{c^2}$ or $pV = \frac{1}{3} Nm \overline{c^2}$ ecf (1) $c_{\text{rms}} = 471$ [m s ⁻¹] (1)	1	1 1		3	2	
		(iv)	Force $= pA = (5.0 \times 10^5) \times 0.04 = 20\,000$ [N]		1		1	1	
	(b)	(i)	Substitution e.g. $p = \frac{5.0 \times 10^5 \times 8.5 \times 10^{-3}}{10.2 \times 10^{-3}}$ (1) ecf on n if $pV = nRT$ used $p = 420$ k[Pa] (1)	1	1		2	2	
		(ii)	$\Delta U = 0$ (1) So by using the first law of thermodynamics $\Delta U = Q - W$ hence $Q = W = 773$ [J] (1)	1	1		2	2	